

The American Community Survey

Three windows onto the same country, one of them closed, and a margin beside every number that has one

84-355 Democracy's Data

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The decennial chapter ended on a short list of things the census does not ask: income, education, language, occupation, commuting, disability, ancestry. Every one of them is asked here.

The American Community Survey is where nearly everything people call “census data” actually comes from. It is published by the Census Bureau, on the Bureau’s website, in the Bureau’s house style — and it is **a survey**, with all the properties the source chapter gave that word. Roughly 3.5 million addresses a year, continuously, forever.

01 Two instruments, side by side

The question	Decennial census	American Community Survey
Who is asked?	Everyone	A sample: about 3.5 million addresses a year
Must you answer?	Yes – required by law, 13 U.S.C. 221	Yes – required by law, the same statute
How often?	Once a decade, as of one day	Continuously, published every year
What comes out?	Counts	Estimates
How many people live on this block?	Yes – this is the thing only it can do	No – the sample is far too thin
What is the median household income here?	It does not ask	Yes – this is what it is for
Is there a margin of error?	No – but noise is added below the state level	Yes – printed beside every estimate
Can it be used to apportion the House?	Yes – this is its constitutional purpose	No – the Constitution requires an actual enumeration

Read the last row first. The ACS cannot apportion the House, and not for any practical reason. The Census Act tells the Bureau it may use sampling except in determining the population for apportionment, and in 1999 the Supreme Court read that section to bar a sampled apportionment count. The Constitution’s “actual Enumeration” clause is the usual shorthand for this, but the ruling was about the statute and the Court did not reach

the constitutional question. That is the legal line between the two instruments, and every other difference follows from which side of it a number was produced on.

Then read the second row. It answers a question this chapter has been inviting since its first sentence. *If the ACS is a survey, why is it here and not with the other surveys?*

Because you have to answer it. Refusing to answer the American Community Survey is punishable under 13 U.S.C. § 221, the same statute that compels the decennial count. The mail package says so on the envelope: *your response is required by law*. Nothing that arrives later in this book can say that. The American National Election Studies asks about eight thousand people whether they would like to participate, and most of them decline; nobody is fined for hanging up on the General Social Survey.

That difference is not a technicality about enforcement. It decides what each instrument can be made to say.

A survey you may refuse has to persuade its sample to stay. So it can ask what you believe, and it cannot promise that the people who answered resemble the people who did not. A survey you may not refuse can be asked of a random three and a half million addresses. It will still return something the Bureau will publish a margin of error for, a stated range for how far off the estimate could be. It can also be run every year forever, because it is funded as an obligation rather than a grant.

Method puts the ACS with the surveys. Where the data came from puts it here. This book is organised by where the data came from. That is why a sample of 3.5 million addresses is in *Counting People*, and a sample of eight thousand is in *Asking People*. Hold the two side by side when you get there: same country, same people, opposite obligation.

02 Why there is more than one window

A single year of sample is thin. Spread 3.5 million addresses over a country of 3,222 counties and most counties get too few to say anything about. So the Bureau pools years, and publishes more than one product from the same survey.

Window	Counties published	Smallest place published	What it is
One-year (2023)	854	63,661	A single year of sample
Five-year (2019–2023)	3,222	43	Five years of sample pooled, averaged over the period

The one-year file covers 854 counties. The five-year file covers all 3,222. For **2,368 counties there is no one-year estimate at all**, which is 73.5% of the country.

That is the answer to why anyone uses the five-year series. It is not a preference, and it is not caution. For nearly three-quarters of American counties it is the only thing published.

The floor is visible in the file. The smallest county with a one-year estimate holds **63,661 people** — just under the 65,000 the Bureau states as its threshold, because the threshold is not applied to the estimate you are reading. A stated rule and a delivered file rarely line up to the person.

03 The window that was closed

There used to be three windows. Between 2007 and 2013 the Bureau also published a **three-year** series for places above 20,000. It was the middle ground between a thin single year and a five-year average reaching a long way into the past.

It was discontinued. You do not have to take that on faith, and this chapter does not assert it: it asks the server.

Year	Series	Still published	HTTP
2011	3_year	Yes	200
2011	5_year	Yes	200
2012	3_year	Yes	200
2012	5_year	Yes	200
2013	3_year	Yes	200
2013	5_year	Yes	200
2014	3_year	No – 404	404
2014	5_year	Yes	200
2015	3_year	No – 404	404
2015	5_year	Yes	200

The three-year directory answers for 2013 and returns **404** for every year after it. The five-year directory answers for all of them. The discontinuation is a fact you can check with a URL, and the last three-year release was the 2013 vintage.

This is worth more than a historical note. A script written in 2014 that pointed at the three-year series did not start returning wrong numbers — it started returning nothing, which is the good case. The bad case is the analyst who quietly substituted the five-year series and did not adjust the claim, because those two files answer different questions.

04 The same county, two populations

For the 854 counties that appear in both files, you can put the two windows side by side. Same agency, same survey, same release, same county.

Quantity	Value
Counties in both files	854
Counties where the two differ by more than 1%	474
Counties where the two differ by more than 5%	68
Median absolute difference	2,119
Median absolute difference, per cent	1.19%
Largest absolute difference, per cent	15.54%

474 of 854 differ by more than one per cent, and 68 differ by more than five. The largest gap is 15.54%.

Neither number is wrong. The one-year figure is 2023. The five-year figure is an average over 2019–2023, which is a different question with a different answer. The trouble is that both are labelled with the same year and quoted interchangeably.

Look at where the gaps are largest:

County FIPS	State	One year 2023	Five year 2019 2023	Difference pct
48257	TX	185,690	160,718	+15.54%
48397	TX	131,307	116,931	+12.29%
48091	TX	193,928	174,552	+11.10%
48291	TX	108,272	97,993	+10.49%
12119	FL	151,565	137,536	+10.20%
13157	GA	88,615	80,640	+9.89%
48367	TX	173,494	158,079	+9.75%
37019	NC	159,964	145,889	+9.65%

Every one of the eight is a fast-growing exurban county, and in **every one the five-year figure is lower**. The direction is not noise. A five-year average includes four years when the place was smaller, so where growth is fast the five-year window *systematically understates the present* — and where a place is shrinking it systematically overstates it.

Choosing a window chooses a finding. If you want to argue a county is booming, the one-year file helps you; if you want to argue it is not, the five-year file does. Both are official.

Before you read on, commit to a view. Every ACS estimate is published with a margin of error beside it. **What fraction of the 3,222 county rows in the ACS table for total population do you think carries a usable margin** — and what would it mean if a Census Bureau table printed no margin at all?

05 Whether a number has a margin depends on which table it came from

Here is the thing that decides how the rest of this chapter had to be written.

Table	County rows	Rows with no margin	Share	Why
B01003, total population	3,222	3,090	95.9%	Controlled to the Population Estimates Program
B19013, median household income	3,222	0	0.0%	Estimated from the sample, like most of the ACS

3,090 of the 3,222 county rows for total population carry no margin of error, or 95.9% of them. What they carry instead is −555555555, which is not a margin of minus half a billion

people. It is the Bureau's code for an estimate that was **controlled**: not estimated freely from the sample at all, but forced to agree with a figure from somewhere else.

That somewhere else is the Population Estimates Program, the subject of the census-pep chapter. So the ACS county population you look up is, in a real sense, not an ACS number.

Median household income is not controlled, and every one of its 3,222 county rows carries a real margin. The lesson is not that one table is better. It is that **"does this number have a margin" is a question about the table, not about the ACS** – and the only way to know is to look. Arithmetic that treats the sentinel as a number produces nonsense quietly. How far the control goes is measured in the access chapter: for a county's total population the one-year estimate is not merely controlled, it is the Population Estimates figure exactly.

06 The margin is not decoration

So, using a table that actually carries margins:

Population of county	Counties	Median income	Median margin	Margin as pct
under 1,000	34	\$62,125	±\$12,772	±20.8%
1,000-4,999	283	61,438 ± 8,669	±14.5%	
5,000-19,999	1,030	59,138 ± 5,566	±9.7%	
20,000-64,999	1,029	61,323 ± 3,833	±6.4%	
65,000-249,999	560	70,454 ± 2,610	±3.6%	
250,000 and over	284	84,572 ± 1,602	±1.9%	

In the largest counties the median margin on household income is **±1.9%** – tight enough to compare places with. In counties under a thousand people it is **±20.8%**, which is wide enough that a county at the national median is statistically indistinguishable from one well above or below it.

And now put that beside the earlier finding. The places with the widest margins are the small ones – which are exactly the places that have no one-year estimate, and for which the five-year file is the only source. **The uncertainty is largest precisely where there is no alternative.**

07 What arrives

The one-year file, opened. Three columns, pipe-delimited: the geography, the estimate, and the margin of error.

```
GEO_ID|B01003_E001|B01003_M001
0100000US|334914896|-555555555
0100089US|1037750|31378
0100091US|2645788|8400
```

```
0100093US|278335|3861
0100094US|1119291|15966
```

The first data row is the United States, and its margin is -555555555. That is not a margin. It is the code the Bureau uses for an estimate that was CONTROLLED rather than freely estimated, and any arithmetic that treats it as a number will produce nonsense quietly. In the five-year file this code appears 3,090 times among the county rows alone.

GEO_ID must be read as text. The county code is characters 10–14, and it keeps its leading zero only if nothing has turned it into a number along the way.

08 What this chapter cannot tell you

One table, one year, one geography. Everything above is total population and median household income, 2023 vintage, at the county level. The ACS publishes thousands of tables down to the block group, and margins at block-group level are far wider than anything shown here. The shape of the finding does not depend on the table: margins widen as places shrink, and windows disagree where places change. The specific numbers do depend on it.

The three-year probe is a check on the server, not on the archive. A 404 means the Bureau does not publish that directory today. Data released between 2007 and 2013 still exists in other forms, and the chapter's claim is only that the series stopped, which is what the file system shows.

Why an estimate was controlled is not visible here. The file records *that* a number was controlled, not the reasoning or the source figure it was controlled to. That the target is the Population Estimates Program is documentation, not something recovered from this file.

Nothing here measures whether the ACS is right. Margins of error describe sampling uncertainty — how much the estimate would move if you drew a different sample. They say nothing about who did not answer, who was not asked, or whether the question meant the same thing to everyone. A narrow margin on a badly measured quantity is still narrow.

09 Sources

- U.S. Census Bureau. *American Community Survey Summary File*, 2023, table-based format: table B01003 (total population), one-year and five-year; table B19013 (median household income), five-year. https://www2.census.gov/programs-surveys/acs/summary_file/2023/table-based-SF/
- U.S. Census Bureau. *American Community Survey Summary File* archive, vintages 2011–2015, used to establish when the three-year series stopped being published. https://www2.census.gov/programs-surveys/acs/summary_file/
- 13 U.S.C. § 221, which makes refusal to answer either instrument an offence, and § 141, which authorises the surveys the Bureau conducts between counts. The obligation

those two sections create is what separates this instrument from every survey later in the book. <https://www.law.cornell.edu/uscode/text/13/221>

- Every published figure for one county's population, and the exact equality between a one-year total and a population estimate, are laid out in this book's access chapter.
- ACS state-to-state migration flows, and what their margins do to a finding, are in the migration chapter.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`compare.csv`, `controlled.csv`, `diverge.csv`, `instrument.csv`, `margins.csv`,
`vintages.csv`, `windows.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `margins.csv` groups counties by size and reports the median margin on median income. Read `median_margin_pct` down the column. At what size of county does the margin stop being a detail and start being the story?
- `diverge.csv` sets the one-year estimate against the five-year estimate for the same county. Sort by `difference_pct`. What kind of county disagrees with itself the most, and which of the two windows would you trust there?
- `controlled.csv` shows that most county rows for total population arrive with no margin at all. Read the `why` column. Is a number with no margin more certain, or just less honest about it?
- `vintages.csv` records which series were published in which year and what the server said. One window is closed. Work out what a researcher lost when it shut.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original source, without any starting files. Please do the following and show your work.

THE SOURCE. U.S. Census Bureau, American Community Survey, 2023

table-based summary files, served plainly from

https://www2.census.gov/programs-surveys/acs/summary_file/2023/table-based-SF/data/

No account or key is needed. Fetch three files:

- `1YRData/acsdt1y2023-b01003.dat` – total population, one-year estimates, about 239 KB
- `5YRData/acsdt5y2023-b01003.dat` – total population, five-year estimates, about 18 MB
- `5YRData/acsdt5y2023-b19013.dat` – median household income, five-year, about the same size

Each is pipe-delimited: a geographic ID, then estimate and margin columns. A GEO_ID starting 0500000US is a county. One warning: a margin of -555555555 is not a margin of minus half a billion. It is a sentinel meaning the estimate was "controlled" to match another Bureau program, so no margin applies. Treat it as a flag, not a number, or your arithmetic will be nonsense.

WHAT TO BUILD.

1. Count the counties in the one-year population file and in the five-year one, and how many counties appear only in the five-year file – places for which no one-year estimate exists at all.
2. For counties in both files, compare the two published estimates of the same quantity: how many differ by more than 1%, and the median absolute difference in people.
3. In each of the two five-year tables, count the county rows whose margin field is the controlled sentinel.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 854 counties in the one-year file and 3,222 in the five-year file, so 2,368 counties have no one-year estimate
- 474 of the shared counties differ by more than 1%, and the median absolute difference is 2,119 people
- 3,090 of the 3,222 population rows carry the sentinel; none of the income rows do

A new vintage appears every year under a new directory name; later numbers mean a different window was measured, not that you erred.